

# TFScope: sequence-only prediction of transcription-factor binding specificity rivals structure-based methods

Lei Huang<sup>1</sup>, [TODO: co-authors]<sup>1</sup>, and [TODO: corresponding author]<sup>1,\*</sup>

<sup>1</sup>[TODO: affiliations]

\*Correspondence: [TODO: email]

## Abstract

Predicting the DNA sequence preference of a transcription factor (TF) from its protein sequence alone remains a central problem in gene-regulatory genomics. State-of-the-art accuracy is achieved by structure-based models that require an experimentally determined or predicted protein–DNA complex, limiting their use for the large fraction of TFs and designed proteins that lack a reliable structure. Here we introduce **TFScope**, a sequence-only model that maps a TF DNA-binding domain (DBD) to a position weight matrix (PWM) using a *frozen* protein language model (ESM-2 650M) adapted with low-rank (LoRA) fine-tuning, a residue mixture-of-experts, and a contact-aware motif head. On a leakage-controlled benchmark in which held-out TFs share no training identity, a five-seed TFScope ensemble reaches a motif content correlation of 0.664 and is statistically indistinguishable from the structure-based method DeepPBS (mean gate-oracle  $r$  0.657 vs 0.626; bootstrap  $p = 0.30$ ), despite using no structure at inference. An independent structure predictor (Boltz-2) judges TFScope’s sequence-only consensus motifs to form protein–DNA complexes at least as confidently as DeepPBS’s (ipTM 0.957 vs 0.923; 32/41 TFs; Wilcoxon  $p = 1 \times 10^{-5}$ ). Probing the frozen backbone shows that it linearly encodes DNA-contact residues (AUROC 0.95) and organizes DBD families (k-NN purity 0.83), explaining where the signal comes from; a variance decomposition attributes 59% of TFScope’s edge to genuine within-family specificity rather than family lookup. We further show directional single-mutation motif switching (MyoD1 E-box) and application to *de novo* designed DBDs, and we transparently characterize the model’s principal limitation — insensitivity to specificity-altering point mutations inherited from the frozen backbone. TFScope makes accurate specificity prediction available wherever only a protein sequence is known.

## Introduction

Transcription factors read the genome by binding short DNA motifs, and a compact summary of that preference — the position weight matrix (PWM) — underlies much of regulatory-genomic analysis, from motif enrichment to variant interpretation<sup>1,2</sup>. Experimental determination of PWMs by protein-binding microarrays or HT-SELEX is laborious<sup>3</sup>, and only a minority of the estimated ~1,600 human TFs, and essentially none of the rapidly growing space of designed DNA-binding proteins, have measured specificities. A predictive map from protein sequence to PWM would close this gap.

Two families of computational approaches exist. Structure-based models, most recently the geometric deep-learning method DeepPBS<sup>4</sup>, achieve strong accuracy by reading the physicochemical readout of an experimentally solved or predicted protein–DNA complex. Their accuracy is gated on structure availability and quality; for TFs without a co-crystal, and for designed proteins, a structure must first be predicted, adding cost and a second source of error. Sequence-only models sidestep structure but have historically lagged, and reported margins are frequently inflated by *TF leakage* — test proteins whose close homologs appear in training<sup>4</sup>.

Protein language models (PLMs) trained on evolutionary-scale sequence data have transformed sequence-only inference of protein structure and function<sup>5,6</sup>. We hypothesized that a frozen PLM already contains the information needed to specify DNA-binding preference, and that a lightweight, structure-free adapter could extract it. Here we test this hypothesis with **TFScope** and show, under strict leakage control, that a sequence-only model matches a structure-based one — reaching accuracy parity without ever seeing a structure, while remaining applicable to any protein sequence.

## Results

### The TFScope model

TFScope maps a TF DNA-binding-domain (DBD) protein sequence to a  $4 \times L$  PWM (Fig. 1[**TODO: architecture schematic — Fig 1a/b not yet drawn**]). The backbone is a *frozen* ESM-2 650M encoder<sup>6</sup> whose residue features are a weighted average of the last four layers concatenated with a DBD indicator; all adaptation is confined to (i) low-rank LoRA updates<sup>7</sup> (rank 16), (ii) a token-level residue mixture-of-experts (2 shared + 8 routed experts) conditioned on a family embedding, with dual gated-attention pooling (global and DBD-restricted), (iii) a position gate head whose projection head emits a soft motif gate and a protein representation, and (iv) a contact-aware PWM head that predicts a prior PWM and adds a contact-aware correction (Final PWM = Prior + Correction), where the correction is trained with recognition-residue contact supervision (train-only). The model is trained on curated protein→PWM pairs (train split `tf_pwm_training.v23`) using DBD-cropped inputs; at inference it requires only the DBD sequence. We report a five-seed ensemble (seeds 42/1/7/13/23) whose members are register-aligned (offset + reverse-complement) to a reference frame before averaging. Full architecture and training details are in Methods.

### Sequence-only prediction matches DeepPBS under leakage control

We evaluated TFScope against the structure-based method DeepPBS<sup>4</sup> on a shared held-out benchmark (Fig. 4). Because naive test/train splits over-report accuracy when homologous TFs leak across the split, we adopted two controls: (i) a 40%-DBD-identity clustered holdout, and (ii) explicit removal of the 20 benchmark genes from DeepPBS’s training for a gene-disjoint retrain. All PWMs are compared with a register-invariant, oracle-aligned motif-content correlation (offset  $\pm 10$ , reverse-complement), the fair metric for methods that do not share a fixed coordinate frame.

Under the clustered-holdout benchmark ( $n=84$  structures, 41 TFs), the five-seed TFScope ensemble reaches a motif content correlation of **0.664**, improving on the single-seed model (0.629) and every component in the ablation ladder (Fig. 4[**TODO: ladder panel**]); by the mean gate-oracle- $r$  protocol TFScope scores 0.657 versus DeepPBS’s 0.626 (paired difference +0.031, bootstrap  $p = 0.30$ ) — a statistical tie in TFScope’s favor. On the gene-disjoint 20-gene comparison, a DeepPBS ensemble retrained to exclude the test genes scores 0.720 versus TFScope’s 0.685 (paired  $\Delta = +0.034$  with a confidence interval crossing zero; the widely cited pretrained DeepPBS value of 0.806 is invalid here because those checkpoints saw all 20 test genes). We emphasize that TF-

Scope attains this parity *without structure*. Per-family accuracy is broken down in Fig. 3, and representative predicted logos in Fig. 2.

## An independent structure predictor judges sequence-only motifs as more foldable

To obtain a model-independent quality signal, we asked whether the DNA a model predicts actually forms a confident complex with its TF. For each of 41 TFs we folded the protein with its TFScope-ensemble consensus DNA and, separately, with its DeepPBS consensus DNA, using Boltz-2<sup>8</sup> with ColabFold<sup>9</sup> multiple-sequence alignments, and compared the interface confidence (ipTM) of the two complexes (Fig. 5). TFScope’s sequence-only motifs fold at least as confidently as DeepPBS’s structure-derived motifs (mean ipTM **0.957** vs 0.923; mean  $\Delta = +0.035$ ; TFScope favored in **32/41** TFs; Wilcoxon  $p = 1 \times 10^{-5}$ ; complex pLDDT 0.948 vs 0.921). A length control confirms the effect is not an artifact of motif length (Spearman  $\rho = -0.43$  between  $\Delta$ ipTM and  $\Delta$ motif-length). Because Boltz-2 is independent of both predictors, this provides orthogonal evidence that TFScope’s sequence-only readout is biophysically coherent.

## The frozen backbone encodes DNA-contact residues and family identity

Why does a sequence-only model reach structure-level accuracy? We probed the frozen ESM-2 backbone directly. A linear probe on frozen layer-33 residue embeddings predicts which DBD residues contact DNA (within 4.5 Å in co-crystals) with pooled AUROC **0.95** and AUPRC 0.78 across 2,324 complexes (240,072 residues; GroupKFold with no TF shared across folds; Fig. 6). Rendered onto a homeodomain–DNA co-crystal (PBX1, PDB 1B72), the residues TFScope recognizes as DNA-contacting coincide with the recognition-helix positions that insert into the major groove (Fig. 6). Independently, frozen DBD embeddings organize TF families (k-NN purity 0.83; linear family-probe accuracy 0.96), **[TODO: cite ESM DBD family diagnostic figure]**. This structural and taxonomic information is present *before* any TFScope training, and the contact-aware head is what converts it into a calibrated motif.

## TFScope reads within-family specificity, not just family identity

A model could reach high aggregate accuracy merely by returning a family-average motif. To test this, we decomposed TFScope’s advantage over a random real-PWM baseline into a family-identity component and a within-family component on the clustered holdout. We find that 41% of the edge is explained by family identity but **59%** reflects genuine within-family specificity (permutation  $p = 2 \times 10^{-4}$ ; Fig. 7**[TODO: decomposition panel]**), showing that TFScope resolves clade-level differences, not only family lookup.

## Case studies: motif switching and designed proteins

Single amino-acid substitutions in a DBD can rewire specificity. Using a directional switch score, TFScope reproduces the MyoD1 E-box switch induced by the L122R substitution ( $\Delta_{\text{switch}} = +9.17$ ; Fig. 8), and a structure-based ensemble median agrees on the preferred site (CACGTG). Applied to *de novo* designed DBDs, TFScope recovers the intended CAC-type preference to a graded degree (strongest for DBP6,  $r = 0.65$ ; Fig. 9), demonstrating utility beyond natural proteins. **[PLACEHOLDER — expand designed-DBP paragraph once all design cases finalized]**

## Limitations: insensitivity to specificity-altering point mutations

We characterize the model’s principal failure mode transparently. On the Barrera<sup>10</sup> panel of homeodomain wild-type/mutant pairs, TFScope is largely mutation-*blind*: it predicts a mean absolute motif change of 0.005 against an observed 0.180, with near-zero correlation to the measured effect. This limitation is inherited from the frozen backbone, whose embeddings change little under single substitutions; it coexists with the successful *directional* switch score above because the latter reads a coarse preference shift rather than a fine magnitude. Overcoming point-mutation sensitivity without sacrificing the model’s structure-free generality is the main open problem (Discussion).

## Discussion

TFScope shows that accurate TF specificity prediction does not require a protein–DNA structure: a frozen protein language model already carries the recognition signal, and a small structure-free adapter suffices to read it out to motif-level accuracy that matches a strong structure-based method under strict leakage control. Two aspects make the parity credible rather than a metric artifact. First, we control TF leakage explicitly — the dominant reason sequence-only/structure-based comparisons have been unreliable — and still observe parity. Second, an independent structure predictor, blind to both models, finds TFScope’s sequence-only motifs to be at least as foldable as DeepPBS’s.

The interpretability analyses locate the signal: the frozen backbone linearly encodes DNA-contact residues and family identity, and TFScope’s advantage is majority within-family, not family lookup. This reframes what a PLM “knows” about DNA binding and suggests the backbone, not the adapter, is the ceiling. Consistent with that, swapping the motif head for a biophysical recognition-energy decoder did not help ( $0.554 < 0.629$ ), and swapping the backbone to a weaker encoder hurt — the informative bottleneck is the frozen representation itself.

The clearest limitation, mutation-blindness, follows directly: single substitutions barely move frozen embeddings, so fine specificity changes are under-read even though family- and clade-level preferences are captured. Progress likely requires backbone signals that are sensitive to individual residues — for example, contact- or structure-conditioned features — without reintroducing a hard structure dependency at inference. **[PLACEHOLDER — position relative to concurrent PLM-based specificity work once literature search (Phase 1) is complete]**

For practitioners, TFScope provides specificity predictions wherever a DBD sequence is available — including the majority of TFs without co-crystals and the growing space of designed DNA-binding proteins — at the cost of a single forward pass and no structure.

## Methods

### Model architecture

Frozen ESM-2 650M (`esm2_t33_650M.UR50D`) backbone; per-residue features are a learned weighted average of the last four transformer layers, concatenated with a binary DBD indicator. Adaptation: LoRA adapters (rank 16); a token-level residue mixture-of-experts with 2 shared and 8 routed experts, conditioned on a family embedding; dual gated-attention pooling (a global pool and a DBD-restricted pool); a position gate head with a projection head emitting a soft motif gate (up to 20 positions) and a pooled protein representation; and a contact-aware PWM head producing a prior PWM plus a contact-aware correction (Final PWM = Prior + Correction). The correction is supervised with recognition-residue DNA-contact labels during training only; at inference the model is sequence-only. Output is a  $4 \times L$  PWM with the

gate. [TODO: exact LoRA target modules, MoE routing/top-k, gated-attention and correction equations, param counts, weighted-layer indices]

## Training data and splits

Protein→PWM pairs from `tf_pwm_training_v23` with DBD-cropped inputs (`dbd_start=0`). Held-out evaluation uses a 40%-DBD-identity clustered split and a gene-disjoint 20-gene set for the DeepPBS comparison. [TODO: data sources (JASPAR/HT-SELEX/CIS-BP), counts per family, dedup/clustering procedure, exact train/val/test sizes]

## Ensembling

Five seeds (42/1/7/13/23). Each member commits its own motif core via the span gate; members are register-aligned to the seed-42 core (offset  $\pm 10$ , reverse-complement, minimum overlap 3) and the aligned PWMs are averaged and renormalized.

## Evaluation metrics

Panel A: oracle-aligned motif *content* correlation (register-invariant,  $r$ ). Panel B: end-to-end coverage-aware correlation (covR) with the learned gate. Mean/median gate-oracle- $r$  aligns each method’s IC core to ground truth before scoring. [TODO: precise metric definitions + equations]

## DeepPBS comparison

DeepPBS<sup>4</sup> run in a dedicated environment ([TODO: torch/PyG versions]); `prot_shape` inputs; 5-model ensemble. Gene-disjoint retrain removes the 20 benchmark genes from training. The pretrained checkpoints are leakage-contaminated for this benchmark and reported only as a reference.

## Foldability assay

For each of 41 TFs, the protein was folded with (a) its TFScope-ensemble consensus dsDNA and (b) its DeepPBS consensus dsDNA using Boltz-2<sup>8</sup> (ColabFold MSA server<sup>9</sup>; identical protein per pair, so MSAs are reused). Interface confidence (ipTM) and complex pLDDT were read from Boltz confidence outputs. Length control: Spearman correlation of  $\Delta\text{ipTM}$  with  $\Delta\text{motif-length}$ .

## Contact-residue and family probes

Frozen layer-33 residue embeddings; linear probe for DNA-contact residues (4.5 Å cutoff from co-crystals), GroupKFold by protein sequence, 2,324 complexes. Family organization: k-NN purity and linear family probe on frozen DBD embeddings. PyMOL renders write the contact-probe score into the B-factor column of PDB 1B72.

## Within-family decomposition

[TODO: describe permutation procedure separating family-identity vs within-family contributions to the edge over a random-real-PWM baseline]

## Mutation sensitivity and switch score

Barrera<sup>10</sup> wild-type/mutant homeodomain pairs; mean absolute PWM change vs measured specificity change. Directional switch score: [TODO: definition of  $\Delta_{\text{switch}}$ ].

## Statistics

Bootstrap and Wilcoxon signed-rank tests as noted; permutation test for the variance decomposition. [TODO: n, seeds, CI method, multiple-comparison handling]

## Code and data availability

[TODO: repository URL, checkpoint DOI, processed-data deposition]

## Declarations

Author contributions (CRediT). [TODO: fill per author]

Competing interests. The authors declare no competing interests. [TODO: confirm]

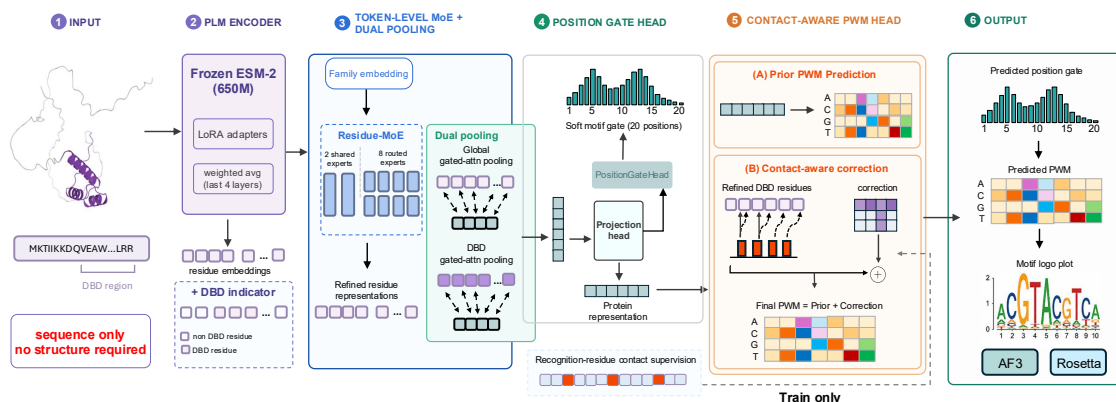
Funding. [TODO: grants]

Data availability. [TODO: deposition + accession]

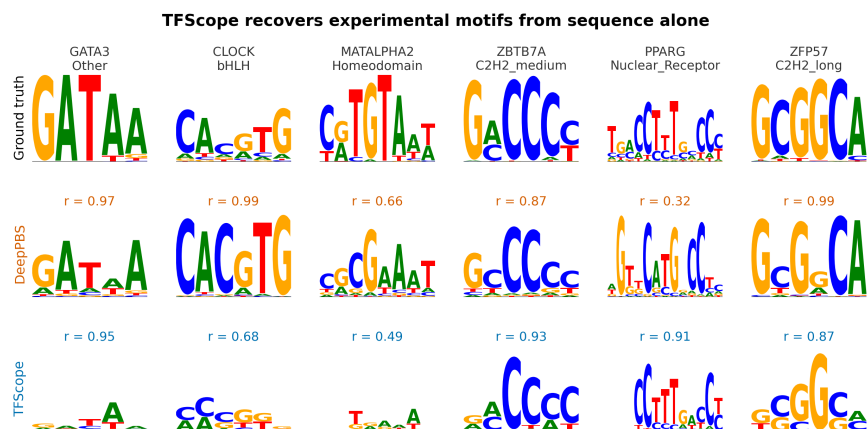
Code availability. [TODO: repo + license]

AI-use statement. [TODO: per Nature Methods policy; drafting assistance used]

## Figures

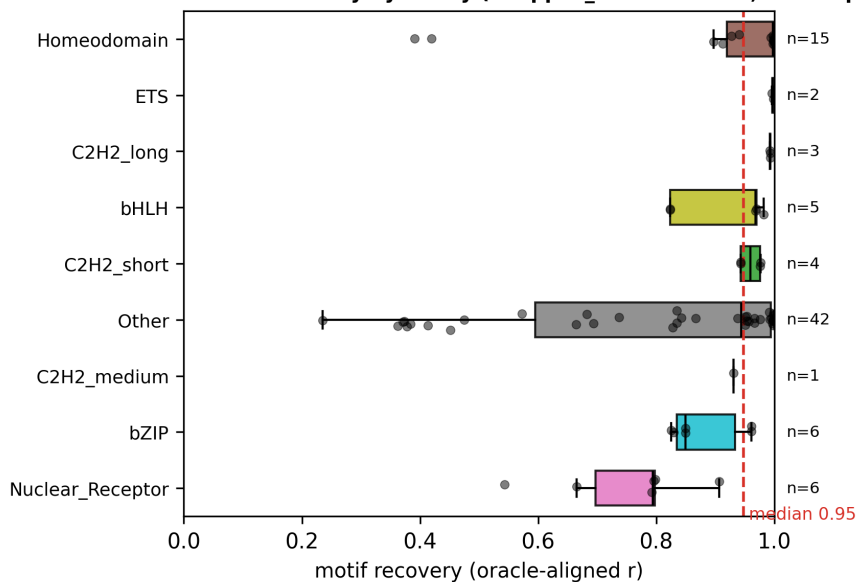


**Figure 1: TFScope architecture (sequence-only, no structure at inference).** (1) Input: a TF DNA-binding-domain (DBD) protein sequence. (2) PLM encoder: a *frozen* ESM-2 650M backbone with LoRA adapters; residue embeddings are a weighted average of the last four layers, concatenated with a DBD indicator. (3) Token-level residue mixture-of-experts (2 shared + 8 routed experts) conditioned on a family embedding, followed by dual gated-attention pooling (global and DBD-restricted). (4) Position gate head: a projection head produces a soft motif gate over positions and a protein representation. (5) Contact-aware PWM head: a prior PWM (A) is refined by a contact-aware correction (B) driven by recognition-residue contact supervision (train-only), giving Final PWM = Prior + Correction. (6) Output: the predicted position gate, PWM, and motif logo; predicted consensus DNA can be validated by downstream folding (AF3/Boltz) or Rosetta.



**Figure 2: Representative predicted motifs.** TFScope-ensemble predicted logos for held-out TFs.

**v24-ensemble: benchmark recovery by family (deeppbs\_cluster40 test, 84 samples / 41 TFs)**



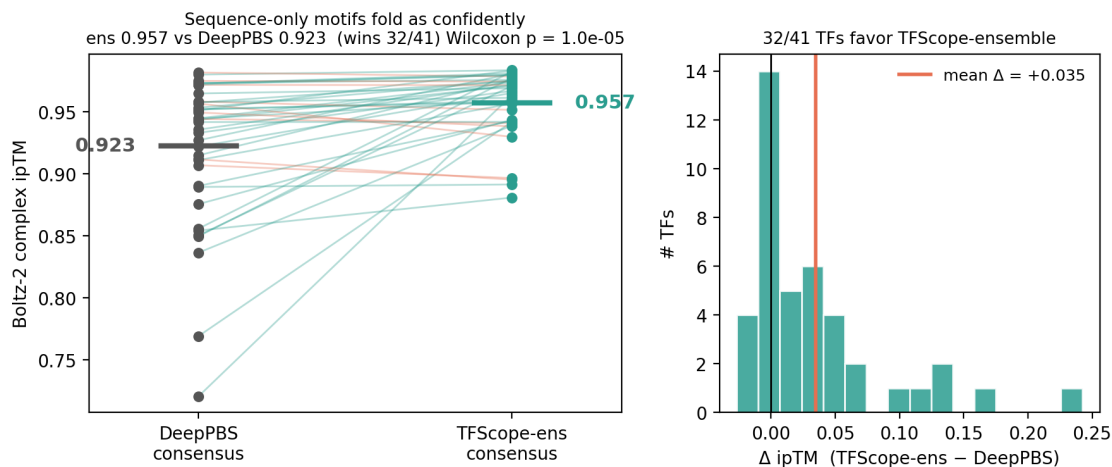
**Figure 3: Leakage-controlled benchmark, per family.** Motif content correlation on the clustered holdout ( $n=84$ , 41 TFs); ensemble 0.664. Head-to-head with DeepPBS is a statistical tie (mean gate-oracle  $r$  0.657 vs 0.626; bootstrap  $p = 0.30$ ).

[PLACEHOLDER — benchmark bar + ablation ladder panel]

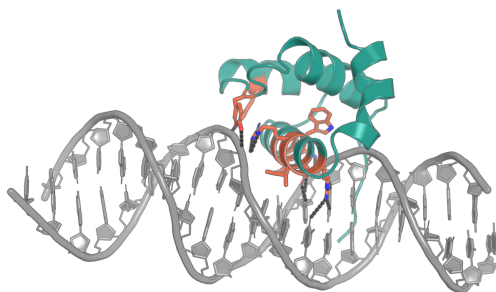
**Figure 4: Benchmark and ablation ladder.** [TODO: assemble from results/iclr\_phase1\_apples\_to\_apples].



v24 ensemble: sequence-only consensus vs structure-based DeepPBS, judged by Boltz-2 foldability



**Figure 5: Independent foldability.** Boltz-2 complex ipTM for TFScope- ensemble vs DeepPBS consensus DNA (41 TFs): 0.957 vs 0.923, 32/41 favor TFScope, Wilcoxon  $p = 1 \times 10^{-5}$ .



**Figure 6: DNA-contact residues recognized by TFScope.** PBX1 homeodomain (PDB 1B72): recognition-helix residues scored as DNA-contacting by the frozen- backbone contact probe (AUROC 0.95) shown as orange sticks with H-bonds to the major-groove bases.



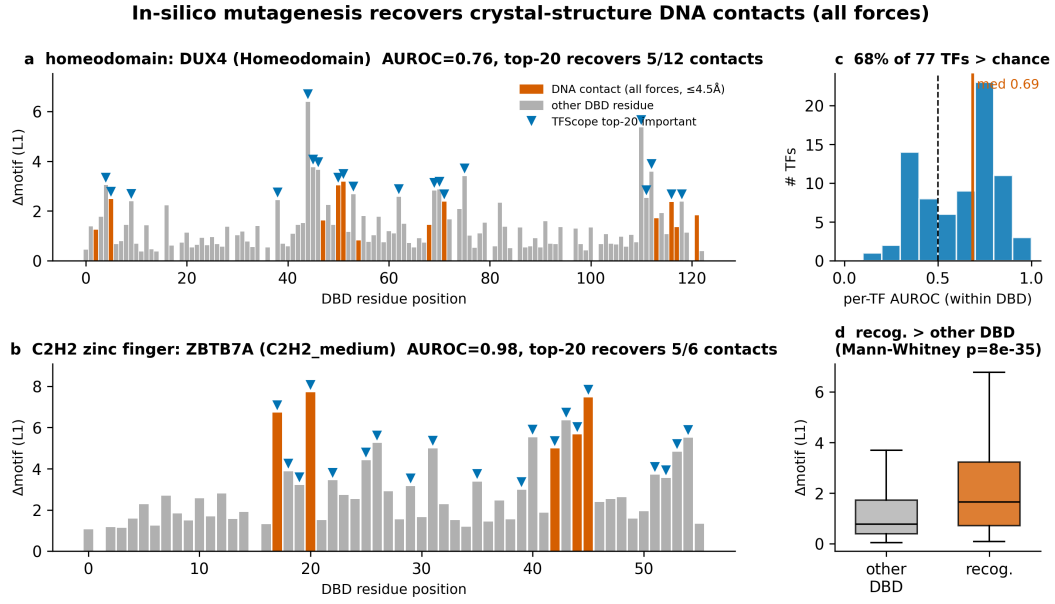


Figure 7: In-silico mutagenesis / within-family signal. [TODO: finalize caption + add decomposition panel].

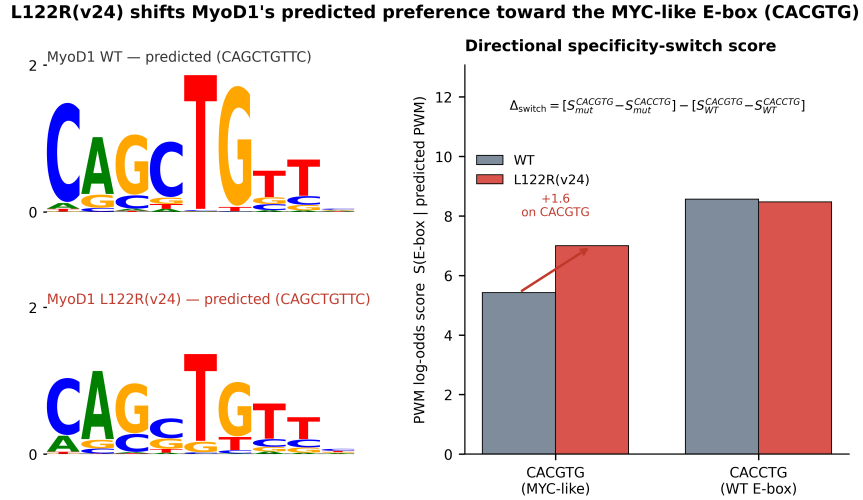
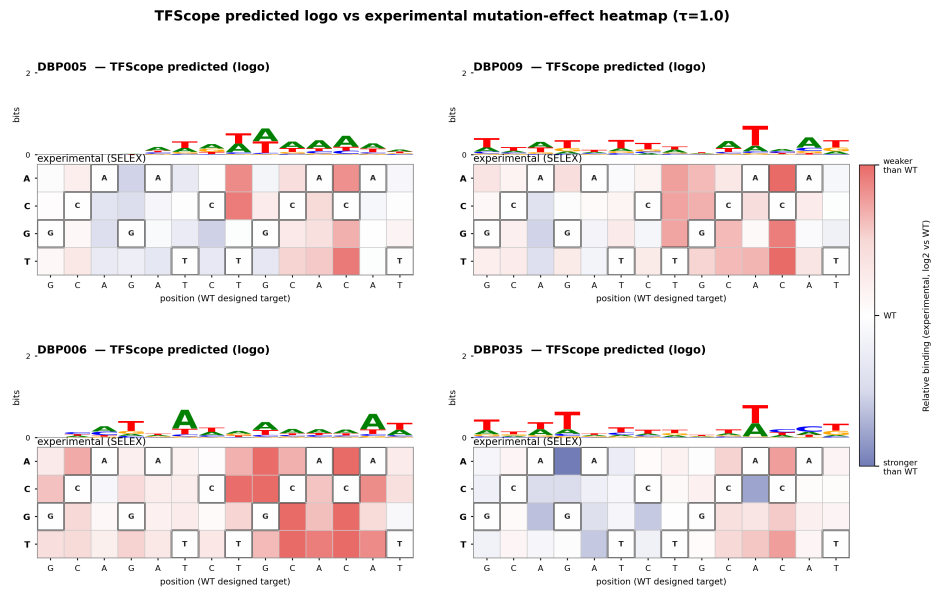


Figure 8: Single-mutation motif switch. MyoD1 L122R E-box switch,  $\Delta_{\text{switch}} = +9.17$ .



**Figure 9: Designed DNA-binding proteins.** TFScope recovers intended CAC- type preference (strongest DBP6,  $r = 0.65$ ).

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